Practice workbook · The final 20-mark program

The last question of Paper 2: a multi-requirement program, usually built on 1D/2D or parallel arrays of test data, sometimes with a menu, subprograms, validation, searching or ordering; 11 functionality marks plus a 9-mark levels band.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-09

1. Question 6 · part whole · 20 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

6 An agricultural college has a herd of dairy cows.

Data collected for the cows is stored in arrays.

The data stored is:

- the name of the breed
- the ease of care rating for that breed (1 is highest and 3 is lowest)
- the number of cows in the herd of that breed
- the volume of milk per day for a cow of that breed.

The college wants to present the data to farmers and to recommend the best breed of cows for them.

The best breed has the highest care rating and the largest volume of milk per day for a cow of that breed.

Open the file **Q06** in the code editor.

Write a program to:

- calculate the daily volume of milk produced by each breed
- add this daily volume to the data structure tbl_dailyVolume
- display a message informing the user what each field holds, such as breed, rating, volume per cow, count and total volume
- display the data for each breed
- calculate and display the total volume of milk produced each day by the herd
- find the recommended breed
- display the recommended breed by name.

Figure 8 shows the output from a functional program.

```
Fields are: Breed, Rating, Volume (cow), Count, Volume (day)
Red Chittagong 1 7.5 6 45.0
Sussex 2 5.7 3 17.1
Dexter 3 11.4 8 91.2
Abondance 2 11.4 7 79.8
Sahiwal 3 22.0 6 132.0
Vorderwald 1 15.2 4 60.8
Ayrshire 2 21.0 3 63.0
Jersey 1 18.3 7 128.1
Randall 2 19.0 3 57.0
Alderney 1 9.0 3 27.0
Carora 3 23.1 4 92.4
Gloucester 2 16.0 7 112.0
Total: 905.4 litres
Recommended breed: Jersey rating: 1 volume 18.3
```

Figure 8

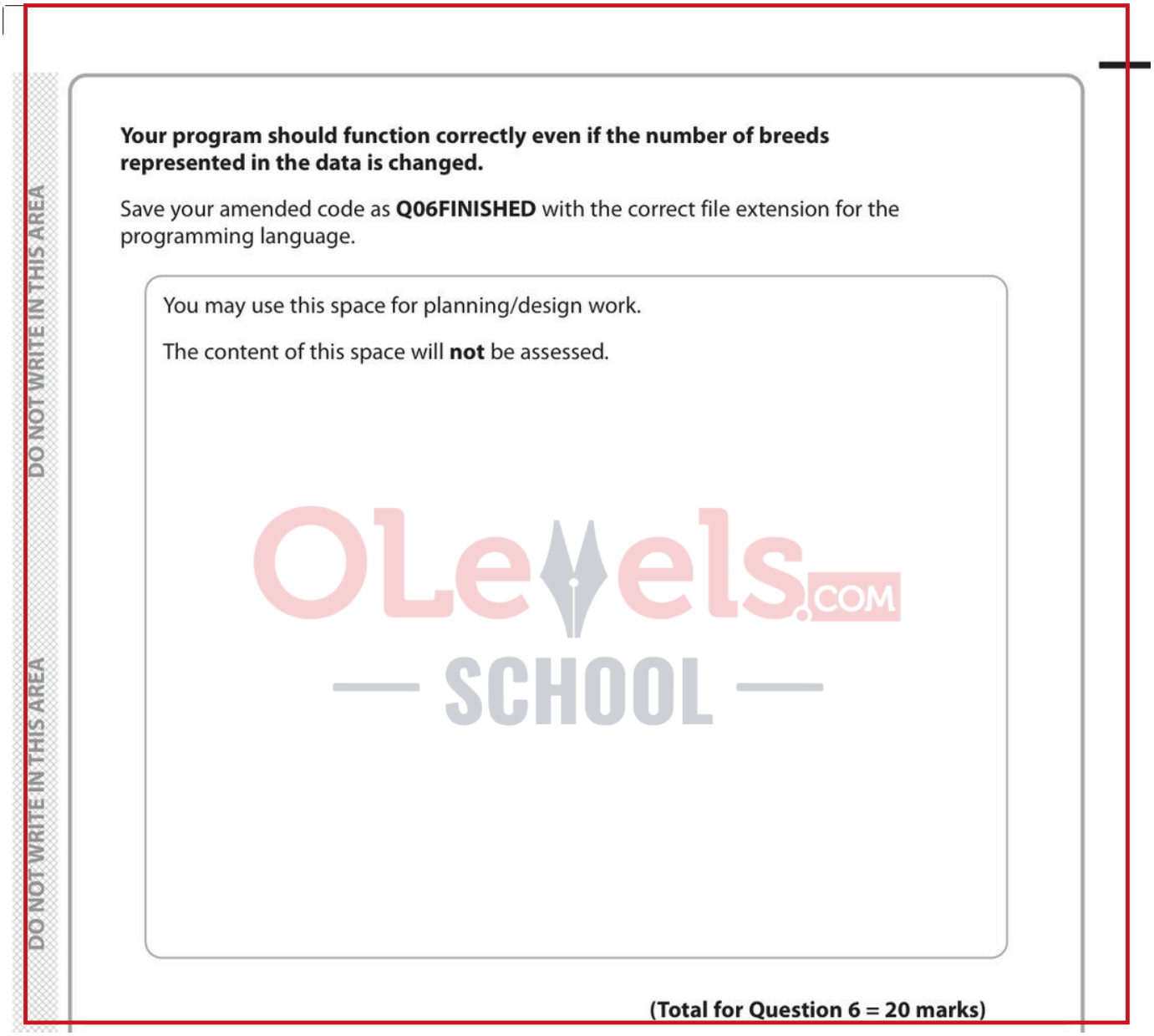
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P 7 2 9 3 8 A 0 1 4 1 6

1. Question 6 · part whole · 20 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.



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Your program should function correctly even if the number of breeds represented in the data is changed.

Save your amended code as **Q06FINISHED** with the correct file extension for the programming language.

You may use this space for planning/design work.

The content of this space will **not** be assessed.

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(Total for Question 6 = 20 marks)

1. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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2. Question 5 · part whole · 20 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.

5 Ria is a school librarian.

She wants a program to analyse pupil use of the library.

She wants to encourage reading by awarding gold, silver and bronze medals to the three pupils who have read the most books.

Test data has been included in the code.

Open **Q05** in the code editor.

Write a program to calculate and display:

- the total number and average number of books pupils have read
- the IDs of pupils who have read fewer than ten books
- the details of the gold, silver and bronze medal winners.

Your program should function correctly even if the number of pupils in the file is changed.

Save your code as **Q05FINISHED** with the correct file extension for the programming language.

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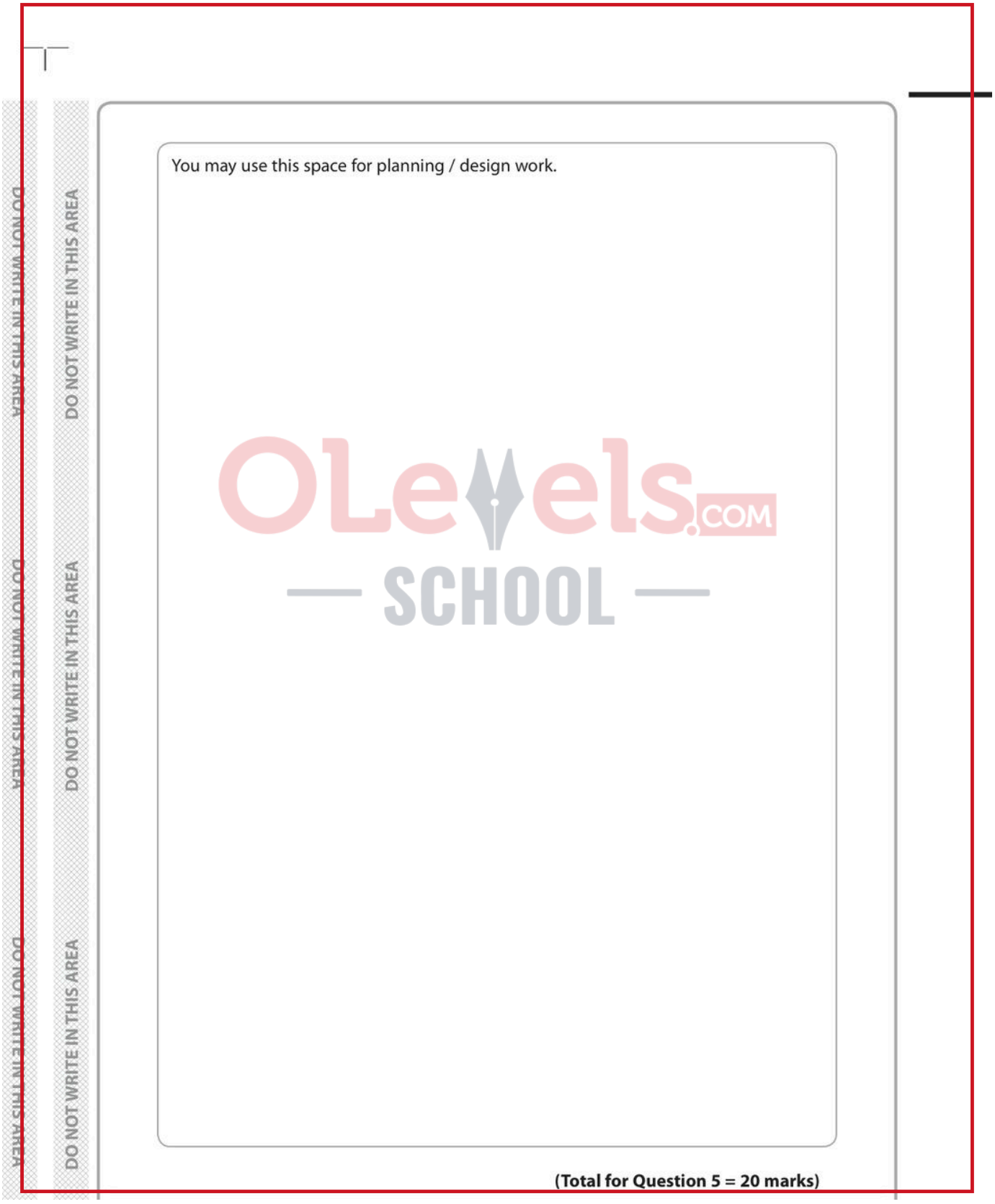
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2. Question 5 · part whole · 20 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.



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(Total for Question 5 = 20 marks)

2. Question 5 · part whole · 20 marks · Working and answer

Working and answer

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3. Question 6 · part whole · 20 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

6 Farshia is the regional manager for an insurance company.

She manages a team of sales staff.

She wants a program to analyse the performance of her team over a number of months.

Test data has been included in the code.

Open **Q06** in the code editor.

Write a program to:

- calculate and display the total sales made by each member of the team
- calculate and display the total sales made by the whole team
- display the first name, last name and the total sales made by the two members of the team with the highest total sales. (Ignore the possibility of two or more members of the team having the same total sales.)

Your program should function correctly even if the number of months or number of members of the team is changed.

Save your code as **Q06FINISHED** with the correct file extension for the programming language.

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3. Question 6 · part whole · 20 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

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(Total for Question 6 = 20 marks)

3. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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4. Question 5 · part whole · 20 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

5 Bianca has started to write a program.

The program is a guessing game about countries and their capital cities.

Open **Q05** in the code editor.

The program already:

- displays this menu

```
                    Menu
----------------------------------------
[1] Add player name
[2] Play guess the capital city
[3] End game
----------------------------------------
Input your menu choice:
```

- asks for the user's menu choice.

Write the program code for the menu choices.

Menu choice [1] Add player name

If a player chooses this option, then they must input a player name.

Menu choice [2] Play guess the capital city

If a player chooses this option, then:

- they must answer five questions
- they select a question to answer by choosing a valid question number
- each question can only be selected once (they must not be able to choose a number more than once)
- the question should be displayed
- they must input the name of the capital city
- if they guess correctly their score must be incremented by 1
- if they guess incorrectly the correct answer must be displayed.

Menu choice [3] End game

If the user chooses this option, then the player name and score should be displayed.

Your program should function correctly even if the number of countries and capital cities in the file is increased.

Save your code as **Q05FINISHED** with the correct file extension for the programming language.

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4. Question 5 · part whole · 20 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

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(Total for Question 5 = 20 marks)

4. Question 5 · part whole · 20 marks · Working and answer

Working and answer

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5. Question 5 · part whole · 20 marks

2021 NOV · Paper 02

Further practice · Attempt part whole only.

5 Ann likes to create programs.

Open **Q05** in the code editor.

It contains an array of words.

Ann wants the program to:

- accept the input of a word, or the number 1 to exit. No validation of the input is required
- display all the words in the array that have the same first letter as the word that has been input
- calculate and display the number of words in the array that begin with the same letter as the word that has been input
- display all the words in the array that contain the word that has been input
- calculate and display
  - the number of words in the array that contain the word that has been input
  - the number of letters in the longest word that contains the word that has been input
  - the number of letters in the shortest word that contains the word that has been input
- display
  - the longest word that contains the word that has been input
  - the shortest word that contains the word that has been input
- repeat until the user inputs the number 1 to exit the program.

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5. Question 5 · part whole · 20 marks

2021 NOV · Paper 02

Further practice · Attempt part whole only.

Figure 4 shows the display when the word **win** is input.

```
Enter a word or 1 to exit: win
----------
wind
wire

2 word(s) begin with w
----------
wind
borrowing
swing

3 word(s) contain win
The longest word has 9 letters
The shortest word has 4 letters
The longest word is borrowing
The shortest word is wind
----------
Enter a word or 1 to exit:
```

Figure 4

Write the code for the program.

Your program should function correctly even if the number of words in the array is changed.

Save your code as **Q05FINISHED** with the correct file extension for the programming language.

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5. Question 5 · part whole · 20 marks

2021 NOV · Paper 02

Further practice · Attempt part whole only.

You may use this space for planning / design work.

The content of this page will **not** be assessed.

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(Total for Question 5 = 20 marks)

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5. Question 5 · part whole · 20 marks · Working and answer

Working and answer

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6. Question 6 · part whole · 20 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.

6 Carlos wants you to create a **guess the animal** game.

Open **Q06** in the code editor.

The code contains an array of animals.

It also contains a function that randomly selects an animal from the array. This is the secret word the user needs to guess.

Carlos wants the program to:

- generate the number of attempts the user has to guess the secret word.
  The maximum number of attempts is the length of the secret word +3. For example, the user has 8 attempts to guess when the secret word is **tiger**
- keep track of letters from incorrect attempts that are in the secret word and those that are not. There should be no duplicated letters
- display a message telling the user:
  - the number of letters in the secret word
  - how many attempts they have left
- force the user to input a word that is the same length as the secret word
- check whether the input word matches the secret word:
  - if the words match then a message that includes the secret word and the number of attempts taken to guess it is displayed
  - if the words do not match then:
    - letters from the attempt that appear in the secret word should be added to the correct letters store
    - letters from the attempt that do not appear in the secret word should be added to the wrong letters store
    - the contents of the correct and wrong letter stores are displayed
- allow the user another attempt until they have guessed the word or have run out of attempts
- display a message telling the user the game is over including the random word if the maximum attempts have been taken and the word has not been guessed.

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6. Question 6 · part whole · 20 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.

Figure 4 shows the contents of the correct and wrong stores after two attempts to guess the secret word **cow**.

Secret word cow			
First attempt		**Second attempt**	
input	dog	**input**	cat
correct store	o	**correct store**	o c
wrong store	d g	**wrong store**	d g a t

Figure 4

Your program should include at least two subprograms that you have written yourself.

You must include comments in the code to explain the logic of your solution.

Save your code as **Q06FINISHED** with the correct file extension for the programming language.

You may use this space for planning/design work.

The content of this space will **not** be assessed.

(Total for Question 6 = 20 marks)

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6. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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7. Question 6 · part whole · 20 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

6 A program stores pairs of words in a two-dimensional array.

Each word in a pair starts with a different letter.

Each pair of words should be in alphabetical order, but some are not.

The last pair in the array is an empty pair.

The program must:

- replace the final blank pair of words with the variables **word1** and **word2**
- display the pair number of each pair, followed by each word in the pair, without punctuation
- display the longer word in the pair, indented
- display any pair found to be not in alphabetical order, in alphabetical order, indented, without punctuation.

Figure 9 shows part of the intended output from a functional program.

```
1 apple banana
      banana
2 wrist leg
      wrist
      leg wrist
3 blue yellow
      yellow
4 speaker keyboard
      keyboard
      keyboard speaker
```

Figure 9

Open the file **Q06** in the code editor.

Write a program to produce the intended output.

You must use the structure and variables given in **Q06** to complete the program.

Your program should function correctly even if the number of pairs in the array is changed.

You should use techniques to make your code easy to read.

Save your amended code as **Q06FINISHED** with the correct file extension for the programming language.

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Turn over ▶

7. Question 6 · part whole · 20 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

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The content of this space will **not** be assessed.

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(Total for Question 6 = 20 marks)

7. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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8. Question 6 · part whole · 20 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

6 Environmental scientists monitor the annual average temperatures of major cities around the world.

City names and their annual average temperatures are stored in two arrays.

- tbl_city stores the city names.
- tbl_temp stores the annual average temperature for each city in either Celsius or Fahrenheit. A temperature of 25° Celsius is stored as **C25** and a temperature of 80° Fahrenheit is stored as **F80**.

Scientists want a program that will enable them to:

- update temperature data already stored
- calculate the overall average annual temperature of all the cities
- view the data in a table as shown in **Figure 10**.

City	Celsius	Fahrenheit
Reggane	39	103
Cairo	28	82
New Delhi	34	93
Muscat	35	95
Athens	29	84
Barcelona	26	79
Havana	29	84
Phoenix	35	95
Brisbane	25	77
Darwin	27	81
Overall Average	30	87

Figure 10

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8. Question 6 · part whole · 20 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

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The program must:

- allow a scientist to input the name of a city and its temperature in either Celsius or Fahrenheit
  - if the city is in the array, its temperature is updated
  - if the city is not in the array, an appropriate error message is given
- display a table with:
  - a header row and dividing line
  - each city and its annual average temperature in both Celsius and Fahrenheit shown as whole numbers
  - the overall average annual temperature for all cities in both Celsius and Fahrenheit shown as whole numbers

The formula to convert from Celsius to Fahrenheit is: F = C x 1.8 + 32

The formula to convert from Fahrenheit to Celsius is: C = (F – 32) ÷ 1.8

Open the file **Q06** in the code editor.

Write the code to produce this program with the intended output.

Include at least one subprogram.

Your program should function correctly even if the number of cities represented in the data is changed.

Use techniques to make your code easy to read.

Save your amended code as **Q06FINISHED** with the correct file extension for the programming language.

(Total for Question 6 = 20 marks)

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8. Question 6 · part whole · 20 marks · Working and answer

Working and answer

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